H2. Which domains and features mark recently duplicated genes?

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Genes in the youngest dS bins (dS <= 0.1, 'recent') are compared with other paralog genes of the same genome and protein-length class. Method and statistics: `scripts/gene_recent_enrich.py` and `scripts/README.md`.

Status: interim. All numbers come from the 7,384-genome interim tables (yn00 dS). They will change when the final tables are published.

H2.1 Recent duplicates are enriched for transposon-derived genes

Hypothesis. Pfam domains named for transposases, integrases or retroelement proteins are over-represented among recent paralogs.

It would count against the hypothesis if: fewer than 40% of the top enriched domains have transposon-related names.

Test. Among the 40 highest-odds-ratio domains in the clean genome set (q < 0.01, odds ratio >= 2, >= 300 focal genes), count those whose Pfam *name* matches a transposon-related pattern. The pattern is my choice. I wrote it after seeing the top-domain list, so it is post hoc and liberal (it also matches names such as Chromo, OTU and Peptidase_C48). The pattern is: `transpos|Tnp|RVT|^rve|rva|MULE|hAT|Dimer_Tnp|Myb_DNA-bind|CxC|KDZ|Plavaka|Tet_JBP|Cas12f1|gag RVT|DDE|Asp_protease|Transposase|Helitron|Chromo|DUF6532|DUF6698|DUF6830|FLYWCH|OTU|Peptidase` Matches are flagged in the table so the reader can audit them. Verdict rule: SUPPORTED if >= 40%.

Result. 19 of 40 (0.47) match.

dom	name	n_genomes	focal_genes	OR	ci_lo	ci_hi	te_name_match
PF:PF27657	RRT15	138	481	22.7	17.1	30.3	False
PF:PF07282	Cas12f1-like_TNB	123	1.81e+03	14.3	13.1	15.7	True
PF:PF12511	DUF3716	214	902	12.1	10.7	13.8	False
PF:PF26935	DUF8286	53	553	8.96	7.83	10.3	False
PF:PF28402	GloB_C	124	364	8.25	7.02	9.7	False
PF:PF12520	DUF3723	266	964	7.75	7	8.58	False
PF:PF13837	Myb_DNA-bind_4	214	966	6.8	6.08	7.61	True

dom	name	n_genomes	focal_genes	OR	ci_lo	ci_hi	te_name_match
PF:PF12013	OrsD	532	2.54e+03	6.79	6.37	7.25	False
PF:PF23042	KOW1_SPT5	246	1.56e+03	6.43	5.84	7.08	False
PF:PF12776	Myb_DNA-bind_3	429	2.62e+03	6.43	6	6.89	True
PF:PF13966	zf-RVT	90	534	6.37	5.61	7.24	True
PF:PF03439	Spt5-NGN	185	1.12e+03	6.32	5.67	7.06	False
PF:PF29418	DUF3723_N	297	910	6.27	5.68	6.91	False
PF:PF12851	Tet_JBP	231	1.41e+03	6.25	5.71	6.84	True
PF:PF18721	CxC6	204	1.73e+03	5.31	4.89	5.76	True
PF:PF18718	CxC5	227	1.96e+03	5.04	4.68	5.44	True
PF:PF20499	DUF6729	122	549	4.89	4.27	5.6	False
PF:PF13352	DUF4100	269	1.43e+03	4.8	4.41	5.22	False
PF:PF24764	rva_4	492	2.62e+03	4.75	4.48	5.05	True
PF:PF27653	SPOK2_N	270	1.21e+03	4.48	4.13	4.87	False
PF:PF02992	Transposase_21	149	527	4.45	3.91	5.06	True
PF:PF26792	DUF8263	179	1.05e+03	4.44	4.03	4.88	False
PF:PF02338	OTU	78	389	4.42	3.7	5.28	True
PF:PF20681	DUF6818	135	409	4.37	3.76	5.09	False
PF:PF18803	CxC2	504	6.72e+03	4.33	4.17	4.51	True
PF:PF28437	DUF8397	106	587	4.29	3.76	4.89	False
PF:PF04434	SWIM	122	374	4.07	3.49	4.74	False
PF:PF18758	KDZ	725	1.3e+04	4.05	3.94	4.16	True
PF:PF13359	DDE_Tnp_4	267	983	4.05	3.68	4.45	True
PF:PF11917	DUF3435	425	1.67e+03	4.01	3.75	4.28	False
PF:PF18717	CxC4	92	542	3.85	3.38	4.39	True
PF:PF18802	CxC1	313	1.76e+03	3.8	3.53	4.09	True
PF:PF18759	Plavaka	899	2.47e+04	3.73	3.65	3.8	True
PF:PF10551	MULE	219	1.2e+03	3.72	3.4	4.08	True
PF:PF20149	DUF6532	714	1.08e+04	3.51	3.42	3.62	True
PF:PF02902	Peptidase_C48	498	2.17e+03	3.37	3.16	3.59	True
PF:PF22594	GTP-eEF1A_C	353	827	3.33	3.03	3.66	False
PF:PF03143	GTP_EFTU_D3	373	866	3.28	2.99	3.59	False
PF:PF29850	DUF8590	311	1.49e+03	3.23	2.99	3.5	False
PF:PF01612	DNA_pol_A_exo1	271	938	3.18	2.91	3.49	False

Verdict: PARTLY SUPPORTED (by domain name only, post hoc pattern). This rests on domain names, which I did not check against Pfam family descriptions. Several top domains are of unknown function. A direct TE test for one family (OrsD, report H3) found no proximity to TE-domain genes.

H2.2 Young paralog genes are shorter and less often secreted

Hypothesis. Recent paralogs include short gene models (fragments, small ORFs), so their median protein length is lower than that of old paralogs and fewer have a signal peptide.

It would count against the hypothesis if: median lengths are equal or higher in the young bins.

Test. Per-genome median protein length of each bin against the OLD (dS > 1.5) bin in the same genome, clean genomes, genomes with >= 10 genes in both bins. Verdict rule: SUPPORTED if the H1 bin median is lower and fewer than 20% of genomes show higher length.

Result.	bin	n_genomes	median_B	median_OLD	frac_genomes_higher	sigp_B	sigp_OLD
Z	2597	260	428	0.0774	0.0328	0.109	0.17
H1	4452	291	431	0.122	0.0336	0.105	0.166

Result.	bin	n_genomes	median_B	median_OLD	frac_genomes_gt_B	sigp_B	sigp_OLD
Y	5188	291	433	0.0715	0.0485	0.104	0.227
MID	6888	320	434	0.034	0.133	0.108	0.64

Verdict: SUPPORTED. Part of the effect can be a detection artefact: short proteins less often keep an alignable gated pair at high dS. The enrichment tests in H2.1 stratify on length for that reason, but the length effect itself is not separated from detection here. The OrsD family is an exception (long proteins, report H3).

H2.3 Recent duplication is concentrated in carbohydrate-active enzyme families

Hypothesis. CAZy families are enriched among recent paralogs.

It would count against the hypothesis if: few or no CAZy families pass q < 0.01 and OR >= 2.

Test. Same test as Pfam, on CAZy family labels (subfamily suffix removed). Verdict rule: NOT SUPPORTED if at most 5 families pass in any set.

Result.	bin	set	families_tested	enriched
recent	all	146	2	GT59, GH64
recent	clean	112	3	GT59, GT33, GT76
recent	clean_Ascomycota	57	2	CBM21, GT4
recent	clean_Basidiomycota	80	1	GT3
identical	all	85	2	GT21, CE9
identical	clean	51	1	GT4
identical	clean_Ascomycota	27	2	GH2, GH128
identical	clean_Basidiomycota	11	0	

Verdict: NOT SUPPORTED. CAZy labels are coarse. A gene's bin depends on sampled pairs. The result applies to the whole-set level, not to individual genomes or lifestyles.

H2.4 The enrichments are robust to clade, protein length and assembly quality

Hypothesis. A real property of young paralogs should hold in both large phyla, with and without length strata, and with or without assembly-suspect genomes.

It would count against the hypothesis if: domain log odds ratios are uncorrelated between Ascomycota and Basidiomycota, or change strongly with length strata or genome filtering.

Test. Spearman correlation of log odds ratios between sets, for domains tested in both. Verdict rule: SUPPORTED if rho > 0.3 between phyla and > 0.7 for the length and genome-filter comparisons.

Result.	comparison	domains	spearman_lnOR	shared_enriched	only_A
Ascomycota vs Basidiomycota (clean), domains tested in both	1194	0.65	63	393	368

Result.	comparison	domains	spearman_lnOR	shared_enriched	only_A
length-stratified vs unstratified (clean)	4381	0.992	1183	44	106
clean vs all genomes	4381	0.887	491	736	363

Verdict: SUPPORTED. Correlation says the ranking is stable, not that each domain is a true biological signal. The first version of this test (strata dropped when the domain was absent from the focal group) inflated nearly all odds ratios; that error was found by checking that 88% of domains passing was implausible.

Figure

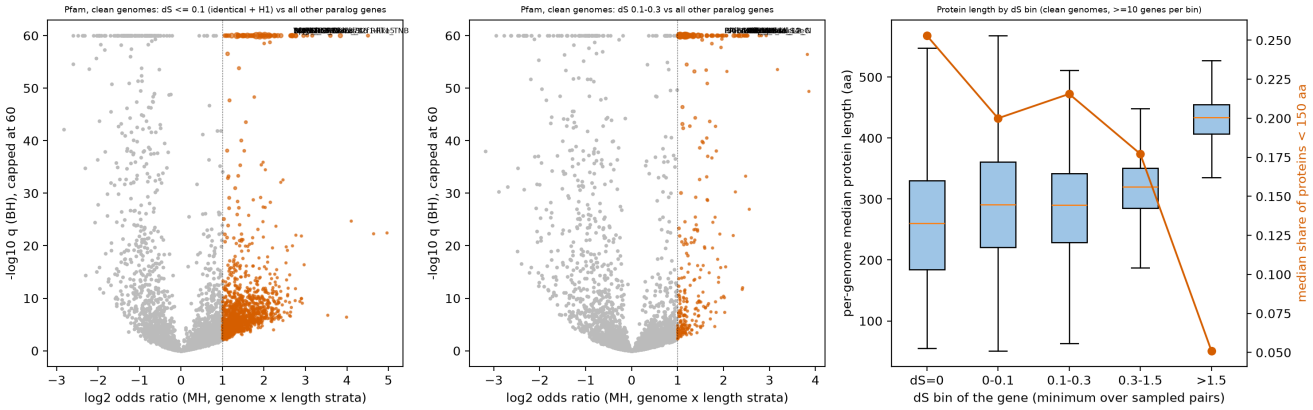


Figure 1: Enrichment volcano and protein length

Reproduce

`scripts/gene_recent_enrich.py`, `scripts/gene_plots.py`, `scripts/make_hypothesis_reports.py`.